EuroSAT Land Classification System

KaggleHacX ’26 — Methodology

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1. Problem Statement

The objective of this project is to classify satellite images into 10 land-use categories using deep learning.
This task is important for applications such as land monitoring, urban planning, and agricultural analysis.

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2. Dataset Description

We used the **EuroSAT dataset**, which contains labeled satellite images across 10 classes:

- AnnualCrop
- Forest
- HerbaceousVegetation
- Highway
- Industrial
- Pasture
- PermanentCrop
- Residential
- River
- SeaLake

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3. Data Preprocessing

To prepare the data for training, the following steps were applied:

- Images resized to **224 × 224 pixels**
- Normalization using ImageNet statistics
- Data augmentation techniques:
    - Random horizontal flip
    - Random rotation

We ensured consistency between training and inference preprocessing, which significantly improved prediction reliability.

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4. Model Selection

We used **ResNet18**, a convolutional neural network pretrained on ImageNet.

Reason for selection:

- Efficient and fast training
- Strong feature extraction capability
- Suitable for transfer learning in limited time

The final fully connected layer was modified to output predictions for 10 classes.

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5. Training Strategy

- Train-validation split: **80-20**
- Loss Function: **CrossEntropy Loss**
- Optimizer: **Adam**
- Learning Rate: **0.001**
- Number of epochs: **3**

Training was performed using GPU acceleration to ensure efficiency.

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6. Validation Approach

Model performance was evaluated using validation accuracy.

Result:

- Achieved approximately **94% validation accuracy**

This indicates strong generalization across different land categories.

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7. Prediction & Inference

For inference:

- Images were preprocessed using the same transformations as training
- Model outputs were converted to probabilities using Softmax
- Final predictions were selected using the highest probability

Additionally, top-3 predictions and confidence scores were provided for better interpretability.

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8. System Implementation

The solution includes:

- A training pipeline (train.py)
- A real-time web application (app.py) using Streamlit
- A prediction system generating submission-ready CSV

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9. Results

The model demonstrated:

- High classification accuracy
- Stable predictions across classes
- Real-time inference capability

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10. Conclusion

This project successfully delivers a complete and efficient solution for satellite image classification.

By leveraging transfer learning and a clean engineering pipeline, we achieved strong performance within limited time constraints.

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11. Future Scope

- Use deeper architectures (ResNet50, EfficientNet)
- Apply advanced augmentation techniques
- Deploy model for real-time satellite monitoring systems

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12. Key Insight

Maintaining consistency between training and inference preprocessing significantly improved model reliability and prediction accuracy.